

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
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I want to write a doctorate on how the design of renewable support schemes changes what gets built. One question has brought me here, and I have met it twice already in my own work: the support an instrument appears to offer and the incentive it creates are different quantities, and the rules decide the distance between them. Energy market design is argued mostly in theory, while Europe has run the experiments for a decade. Reading those experiments is work for an applied microeconomist.

My training is in economics at the University of Cologne, where the energy specialisation is two modules running from the physical foundations of energy systems to European regulation and market intervention. The second, *Energy Markets and Regulation* with Jun.-Prof. Dr. Oliver Ruhnau, goes into grids, interconnectors and the integrated European electricity market, and I completed it with 1.3 where 1.0 is the maximum. Those modules are why an auction rule reads to me as a physical and regulatory constraint as much as an economic one.

My quantitative preparation is the strongest part of my record: 1.0 in mathematics for economists, 1.3 in econometrics, 1.0 in bachelor mathematics and statistics, and Jehle and Reny in my own time this year. My thesis, graded 1.0 and worth 30 of the degree's 120 ECTS, put it to work on a national scholarship programme in Brazil, on a panel I built from administrative records, five million linked employer-employee observations across 558 microregions, 2005 to 2019, in R through BigQuery. Callaway and Sant'Anna (2021) with doubly-robust inference gives a wage gain of 3.04 per cent in low-exposure regions, distinguishable from zero at the 10 per cent level, on a non-monotonic dose-response that two-way fixed effects reports with the opposite sign. What it settled is that exposure intensity, rather than treatment status, is where the answer sits.

The two directions I would take up are de-risking green investment and multi-technology auction design. My final seminar paper, a research design rather than a set of estimates, sits inside the first. It defines the effective EU ETS carbon cost as the allowance price attenuated by the sector-year free-allocation share, and shows that attenuation is not lump-sum, because activity thresholds and closure provisions tie allocation to production. That is the mechanism your first direction names, and it reaches guarantees, contracts for difference and public insurance. The second direction has the cleaner empirical footing and I would start there. Since the 2016 state aid framework member states have arrived at technology-neutral, technology-specific and mixed designs at different dates. That is staggered adoption of a design rather than of a policy. The AURES II database records the design and outcome of every renewable auction held in the EU and UK, 2011 to 2021, so the panel exists. I would estimate how a move to joint bidding shifts the awarded technology portfolio and the rent per megawatt hour against not-yet-treated member states, and test whether design changes were timed to expected cost falls. Germany's joint auction of April 2018 motivates it: all 200 MW went to solar at 46.7 euros per megawatt hour on average, some eight per cent above the solar-only auction two months before.

Three people at CORE are why I want to do that there. Your work with Yueting Yu on discriminatory versus uniform-price auctions asks how short-run bidding maps into long-run investment, which is the rent question inside the second direction. Quentin Lété works on market design, capacity investment and risk, which is the first. ENERPOL, with Jean Hindriks and François Maniquet, studies who carries the cost of the transition, which is what my thesis estimates. What I would bring to a group that argues these questions formally is their empirical counterpart. In ten years I want to be producing the evidence that European support-scheme reform is judged on. Getting there needs graduate training in the theory this group works in and four years of supervised research on a question I already care about. This position is both.

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